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USER INTERFACE AND LOCK FEATURES FOR POSITIONING MULTIPLE COMPONENTS WITHIN A BODY

Abstract

Disclosed embodiments include apparatuses, systems, and methods for positioning electrodes. In an illustrative embodiment, an apparatus includes a primary electrode defining a lumen, an elongated secondary electrode slidably receivable within the lumen, and a sheath configured to slidably receive the electrodes and to convey the electrodes toward a target region. A housing is coupled with the sheath and movably mounted to motivate the sheath relative to the target region. A primary actuator is coupled with the primary electrode and slidably coupled with the housing to motivate the primary electrode relative to the sheath. A secondary actuator is coupled with the secondary electrode and movably coupled with the primary actuator to be slidable with the electrodes in concert. The secondary actuator is rotatable independently of the primary actuator to travel along a helical path to motivate the secondary electrode to move relative to the target region independently of the primary electrode.

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Background/Summary

PRIORITY CLAIM [0001] The present application is a Continuation of U.S. patent application Ser. No. 17/115,902 filed Dec. 9, 2020 and entitled “USER INTERFACE AND LOCK FEATURES FOR POSITIONING MULTIPLE COMPONENTS WITHIN A BODY,” which claims the priority and benefit of U.S. Provisional Patent Application Ser. No. 62/945,825 filed Dec. 9, 2019 and entitled “USER INTERFACE AND LOCK FEATURES FOR POSITIONING MULTIPLE COMPONENTS WITHIN A BODY,” U.S. Provisional Patent Application Ser. No. 62/945,836 filed Dec. 9, 2019 and entitled “HELICAL GUIDE CHANNEL WITH VARIABLE PITCH,” and U.S. Provisional Patent Application Ser. No. 62/945,843 filed Dec. 9, 2019 and entitled “SLIDABLE COUPLING TO CONNECT DEVICES.”

FIELD

[0002] The present disclosure relates to a user interface and lock features for positioning multiple components within a body.

BACKGROUND

[0003] The statements in this section merely provide background information related to the present disclosure and may not constitute prior art.

[0004] Inserting and manipulating thin elements within living bodies or other objects allows for ever-improving types of analysis, diagnosis, and treatment of those bodies or objects with minimally invasive techniques. By way of two examples, endoscopic imaging and catherization treatments have enabled evaluation and treatment of numerous internal lesions without invasive surgery.

[0005] Electrosurgical techniques also provide for minimally invasive therapies by selectively applying electrical current to selected tissues. Electrosurgical techniques involve inserting one or more electrodes through an orifice or a small incision and then extending the one or more electrodes to a desired location within a body of a patient. A radio frequency (“RF”) electric current is then applied to the electrodes to coagulate, ablate, or otherwise treat tissue at that location. Monopolar electrosurgical instruments involve the insertion of one electrode that electrically interacts with a second electrode that is electrically connected to the body of the patient. A bipolar

electrosurgical instrument involves the deploying of two electrodes at the location within the body of the patient where treatment is to be administered.

[0006] Positioning one or two electrodes at the desired location in a patient's body is an important part of electrosurgical treatments. Moving and holding electrodes in place, particularly when more than one electrode has to be moved or held independently of another electrode, may present a challenge for medical personnel directing the treatment. Further, because positioning one or more electrodes in place may involve following a particular sequence of steps in positioning the electrodes, assisting an operator in properly following the sequence also may be important.

SUMMARY

[0007] Disclosed embodiments include: apparatuses, systems, and methods for controlling the movement of multiple components within a body; apparatuses, systems, and methods for motivating elongated implements using a rotating actuator guided by a helical path of varying pitch; and apparatuses, systems, and methods for coupling a device, such as a user interface for controlling movement of multiple components within a body, to another device.

[0008] In an illustrative embodiment, an apparatus includes an elongated primary electrode defining a lumen therein, an elongated secondary electrode slidably receivable within the lumen, and a sheath configured to slidably receive the primary electrode therein, where the sheath is further configured to convey the primary electrode and the secondary electrode to a target region. A housing is operably coupled with the sheath and movably mounted to slidably motivate the sheath relative to the target region. A primary actuator is operably coupled with the primary electrode and slidably coupled with the housing to motivate the primary electrode relative to the sheath. A secondary actuator is operably coupled with the secondary electrode and movably coupled with the primary actuator to be slidable with the primary actuator to motivate the secondary electrode in concert with the primary electrode. The secondary actuator is rotatable independently of the primary actuator to travel along a helical path to motivate the secondary electrode to move relative to the target region independently of the primary electrode.

[0009] In another illustrative embodiment, a system for treating tissue at a target region includes an electrical power source configured to selectively provide electrical power between a first pole and a second pole via a two-pole electrical cable. An electrode control apparatus includes an elongated primary electrode defining a lumen therein, an elongated secondary electrode slidably receivable within the lumen, and a sheath configured to slidably receive the primary electrode therein, where the sheath is further configured to convey the primary electrode and the secondary electrode to a target region. A housing is operably coupled with the sheath and movably mounted to slidably motivate the sheath relative to the target region. A primary actuator is operably coupled with the primary electrode and slidably coupled with the housing to motivate the primary electrode relative to the sheath. A secondary actuator is operably coupled with the secondary electrode and movably coupled with the primary actuator to be slidable with the primary actuator to motivate the secondary electrode in concert with the primary electrode. The secondary actuator is rotatable independently of the primary actuator to travel along a helical path to motivate the secondary electrode to move relative to the target region independently of the primary electrode.

[0010] In a further illustrative embodiment, a method includes moving a distal end of a sheath that contains a primary electrode and a secondary electrode adjacent to a target region. A primary actuator operably coupled with the primary electrode and a secondary actuator operably coupled to the secondary electrode and movably engaged with the primary actuator are slid to a first position to motivate distal ends of the primary electrode and the secondary electrode relative to the target region. The secondary actuator is rotated relative to the primary actuator to cause the secondary actuator to travel independently of the primary actuator along a helical path to a second position to motivate the distal end of the secondary electrode to move independently of the primary electrode relative to the target region.

[0011] In an additional illustrative embodiment, an apparatus includes an elongated implement

movable along an axis. A rotatable actuator is operably coupled with a proximal end of the implement to motivate the implement to move along the axis in response to rotation of the rotatable actuator. A guide is operably coupled with rotatable actuator, wherein the guide defines a generally helical path around the axis to direct movement of a rotatable actuator, and wherein a pitch of the helical path is varied to reduce a distance of travel of the actuator along the axis per unit of rotation of the actuator.

[0012] In another additional illustrative embodiment, a system includes an elongated primary electrode defining a lumen therein. An elongated secondary electrode is slidably receivable within the lumen. A sheath is configured to slidably receive the primary electrode therein, the sheath being further configured to convey the primary electrode and the secondary electrode toward a target region. A housing is operably coupled with the sheath and movably mounted to slidably motivate the sheath relative to the target region. A primary actuator is operably coupled with the primary electrode and slidably coupled with the housing to motivate the primary electrode to slide relative to the sheath along an axis. The primary actuator includes a guide defining a generally helical path, wherein a pitch of the helical path is varied to reduce movement of a guide member relative to the axis per unit of rotation of the guide member around the helical path. A secondary actuator is operably coupled with the secondary electrode and rotatably received within the guide of the primary actuator. The secondary actuator supports the guide member that is configured to engage the helical path. The secondary actuator is rotatable relative to the primary actuator to motivate the secondary electrode to move relative to the primary electrode.

[0013] In a further additional illustrative embodiment, a method includes coupling an elongated implement at a proximal end thereof to an actuator that is movable along an axis. The implement is motivated by rotatably moving the actuator through a generally helical path around the axis, where the helical path has a pitch that is varied to change a distance traveled by the actuator along the axis per unit of rotation of the actuator.

[0014] In another additional embodiment, a locking body defines an opening with a first section having a first width and a second section having a second width that is smaller than the first width, where the locking body is slidably mountable on one of a first device that supports a first coupling and a second device that supports a second coupling. One of the first and second couplings is configured to support thereon a flange having a flange width that is smaller than the first width and larger than the second width. A slidable mounting mechanism is configured to slidably secure the locking body on one of the first device and the second device. The slidable mounting mechanism is further configured to enable the locking body to slide between an open position, in which the first section is positionable to enable the first coupling to be inserted into the second coupling to form a connection, and a closed position, in which an edge of the locking body around the second section abuts the flange such that the coupling that supports the flange is prevented from being withdrawn from the connection

[0015] In another additional illustrative embodiment, a system includes an elongated primary electrode defining a lumen therein. An elongated secondary electrode is slidably received within the lumen. A sheath slidably receives the primary electrode and is configured to convey the primary electrode and the secondary electrode toward a target region. A housing is operably coupled with the sheath and is movably mounted to slidably motivate the sheath relative to the target region. A primary actuator is operably coupled with the primary electrode and is movably coupled with the housing to motivate the primary electrode relative to the sheath. A secondary actuator is operably coupled with the second electrode and is movably coupled with the primary actuator, where the secondary actuator is separately movable relative to the primary actuator to motivate the secondary electrode to move relative to the primary electrode. A first coupling is supported by the housing and configured to engage a second coupling supporting a flange having a flange width, where the second coupling extends from a device through which the sheath and the electrodes will be conveyed to the target region. A locking body defines an opening having a first section having a

first width larger than the flange width and a second section having a second width that is smaller than the flange width. A slidable mounting mechanism is configured to slidably secure the locking body to the housing. The slidable mounting mechanism is further configured to enable the locking body to slide between an open position, in which the first section is positionable to enable the first coupling to insertably receive the second coupling to form a connection, and a closed position, in which an edge of the locking body around the second section abuts the flange such that the coupling that supports the flange is prevented from being withdrawn from the connection.

[0016] In a further additional illustrative embodiment, a method includes positioning a locking body into an open position, where the locking body defines an opening with a first section having a first width and a second section having a second width that is smaller than the first width. The locking body is slidably mounted on one of a first device that supports a first coupling and a second device that supports a second coupling. The first section is disposed between the first coupling and the second coupling when the locking body is positioned into the open position. A connection is formed by inserting the first coupling within the second coupling such that one of the first and second couplings supports a flange having a flange width that is smaller than the first width and larger than the second width. The locking body is repositioned into a closed position in which an edge of the locking body around the second section abuts the flange to prevent the flange from being withdrawn from the connection.

[0017] Further features, advantages, and areas of applicability will become apparent from the description provided herein. It should be understood that the description and specific examples are intended for purposes of illustration only and are not intended to limit the scope of the present disclosure.

Description

DRAWINGS

[0018] The drawings described herein are for illustration purposes only and are not intended to limit the scope of the present disclosure in any way. The components in the figures are not necessarily to scale, with emphasis instead being placed upon illustrating the principles of the disclosed embodiments. In the drawings:

[0019] FIG. 1 is a block diagram in partial schematic form of an illustrative system for treating tissue;

[0020] FIGS. 2-5 are schematic diagrams of positioning of distal ends of a sheath, primary electrode, and secondary electrode relative to a target region;

[0021] FIGS. 6A and 7A are schematic diagrams of moving a sheath actuator to position a sheath relative to the target region;

[0022] FIGS. 6B and 7B are schematic diagrams of distal ends of the sheath, a primary electrode, and a secondary electrode relative to the target region corresponding to positions of the sheath actuator of FIGS. 6A and 7A, respectively;

[0023] FIG. 8 is a side view of an illustrative sheath actuator and a sheath lock;

[0024] FIG. 9 is a cutaway view of the sheath actuator and sheath lock of FIG. 8;

[0025] FIG. 10 is a side view of an embodiment of a user interface for positioning components relative to the target region;

[0026] FIG. 11 is an exploded view of the user interface of FIG. 10;

[0027] FIGS. 12A, 13A, 14A, 15A, 16A, 17A, 18A, 19A, 20A, and 21A are side views of an embodiment of the user interface of FIG. 10 being manipulated to position multiple components relative to the target region;

[0028] FIGS. 12B, 13B, 14B, 15B, 16B, 17B, 18B, 19B, 20B, and 21B are schematic diagrams of distal ends of the sheath, the primary electrode, and the secondary electrode relative to the target

region corresponding to positions of the user interface of FIGS. **12A**, **13A**, **14A**, **15A**, **16A**, **17A**, **18A**, **19A**, **20A**, and **21A**, respectively;

[0029] FIG. **22** is a side view of a guide sleeve defining a helical channel of varying pitch for guiding a rotatable actuator;

[0030] FIG. **23** is a side view of sections of the guide sleeve of FIG. **22**;

[0031] FIGS. **24** and **25** are side views of a wire having different cross-sections along its length;

[0032] FIG. **26** is a cross-sectional view of the wire of FIGS. **24** and **25**;

[0033] FIG. **27** is an exploded view of a coupler for joining together devices;

[0034] FIG. **28** is a side view of a locking body of the coupler of FIG. **27**;

[0035] FIG. **29** is a flow diagram of an illustrative method of positioning components using a user interface;

[0036] FIG. **30** is a flow diagram of an illustrative method of motivating an implement using a rotating actuator guided by a helical path of varying pitch; and

[0037] FIG. **31** is a flow diagram of an illustrative method of coupling together devices with a slidably-mounted locking body.

DETAILED DESCRIPTION

[0038] The following description is merely illustrative in nature and is not intended to limit the present disclosure, application, or uses. It will be noted that the first digit of three-digit reference numbers and the first two digits of four-digit reference numbers correspond to the first digit of one-digit figure numbers and the first two digits of two-digit figure numbers, respectively, in which the element first appears.

[0039] The following description explains, by way of illustration only and not of limitation, various embodiments of user interfaces to position electrodes for electrosurgical apparatuses, as well as systems including such user interfaces and methods of using the same. As will be described in detail below, electrosurgical techniques position first and second electrodes in a target region where electrical treatment, such as ablative treatment, is to be applied. For a specific example, the user interfaces and methods of their use may be used for ablating and/or coagulating tissue, removing lesions, and for performing other medical procedures within a lung.

[0040] It will be appreciated that various embodiments of user interfaces described herein may help to simplify the process of positioning the electrodes and holding the electrodes in place. As will be described below, various embodiments of the user interface accomplish the selective positioning and locking in place of the electrodes by depressing a release, sliding one actuator, and rotating another.

[0041] Referring to FIG. **1**, a system **100** is provided for treating tissue at a target region of a patient (not shown in FIG. **1**). The system **100** may be a bipolar or monopolar radio frequency (RF) system, as desired, for treating tissue in a patient. However, various embodiments described herein are configured to position two electrodes at the target region to support implementation of a bipolar treatment system, thereby allowing for electric current to be selectively passed through a particular target region in a patient. Specifically, the system **100** may be employed for coagulation and/or ablation of soft tissue during percutaneous and/or endoscopic surgical procedures, such as, for example, bronchoscopic surgical procedures for partial and/or complete ablation of cancerous and/or noncancerous organ lesions. As will be further described, the tissue is treated by positioning one or more electrodes proximate the tissue to be treated and passing an electrical current through the tissue.

[0042] In some embodiments, the system **100** includes a user interface **102**, an electrosurgical radio frequency (RF) generator operating as a switchable current source **114**, an infusion pump **116**, and an electrosurgical instrument or device **118**, such as, without limitation, a bronchoscope or any other electrosurgical or endoscopic instrument as desired for a particular application. The user interface **102** may be joined with the electrosurgical apparatus **118** with a coupler **150**. The electrosurgical apparatus **118** may be used to convey electrodes (not shown in FIG. **1**) through a

sheath **103** where the user interface **102** may be used to manipulate positions of the electrodes at the target region.

[0043] The user interface **102** electrically communicates with the switchable current source **114** through an electrical conductor **130**. In some embodiments, the electrical conductor **130** is connected to a bipolar outlet **131** on the switchable current source **114** when the system is operated in a bipolar mode. The electrical conductor **130** may be coupled with the outlet **131** using an electrical connector **134** configured to electrically engage the outlet **131**. The electrical conductor **130** may be removably or fixably coupled to the user interface **102**, where a flexible electrical coupling (not shown in FIG. **1**) associated with the user interface **102** electrically couples the current to the electrodes, as further described below with reference to FIG. **11**. In some other embodiments, the system **100** can be operated in a monopolar mode when the electrical conductor **130** is connected to a secondary outlet **133** with an adapter (not shown in FIG. **1**).

[0044] The user interface **102** is further connected to the infusion pump **116** with a tube **132** that facilitates the flow of a conductive fluid, such as saline solution, from the infusion pump **116** to the user interface **101**. As also described below with reference to FIG. **11**, the user interface **102** may include a flexible fluid coupling (not shown in FIG. **1**) that receives the flow of conductive fluid from the infusion pump **116** and delivers the conductive fluid to an interior of a primary electrode where it can be delivered to the target region.

[0045] The switchable current source **114** may be operated with the use of a foot operated unit **120** electrically connected to the switchable current source **114**. The foot operated unit **120** may include a pedal **122** that directs the switchable current source **114** to apply an electrical current to one or more electrodes to cut, ablate, or otherwise treat tissue and a pedal **124** that instructs the switchable current source **114** to apply a lower electrical current to the one or more electrodes to coagulate tissue.

[0046] In various embodiments the electrosurgical apparatus **118** includes an insertion tube **119** that permits insertion of the sheath **103** into a body (not shown) through an orifice or an incision. A distal end **105** of the sheath **103** is delivered to a target region where treatment is to be administered. The sheath **103** contains and conveys the electrodes (not shown) to a desired treatment location. Positioning of the distal end **105** of the sheath **103** and the distal ends of the electrodes (not shown in FIG. **1**) may be controlled by the user interface **102** received by the electrosurgical apparatus **118** as further described below with reference to FIGS. **6A-21B**.

[0047] Referring to FIGS. **2-5**, distal ends of components used to administer treatment are positioned relative to a target region **202** using various embodiments of a user interface **102**. The target region **202**, may include a lesion or any portion of tissue to be treated within a body. Various embodiments of the user interface **102** described below are capable of positioning the components as described with reference to FIGS. **2-5** and as further described with reference to FIGS. **6A-21B**. The description of FIGS. **2-5** is provided as a baseline to describe an application with which various embodiments of the user interface **102** may be used to deploy these components.

[0048] In various embodiments, a secondary electrode **211** is slidably received within a primary electrode **207**, and the primary electrode **207** is slidably received within a sheath **203**. Components contained within other components are represented with dashed lines in FIGS. **2-5**. In various embodiments, the primary electrode **207** is in the form of a needle, with the distal end **209** being configured to pierce tissue, such as tissue comprising the target region **202**. Piercing the tissue at the target region **202** with the primary electrode facilitates positioning the distal end **209** of the primary electrode **207** at a desired position and also facilitates conveying the secondary electrode **211** to a desired location. In various embodiments, until a user interface is manipulated to separately move the secondary electrode **211**, the primary electrode **207** and the secondary electrode **211** move in concert, at a same time and through a same distance, with each other and with the sheath **203**.

[0049] Referring to FIG. **2**, the sheath **103**, the primary electrode **207**, and the secondary electrode

211 are positioned at an initial position near the target region **202**. The sheath **103** and the electrodes **207** and **211** received therein may be conveyed to this location through the use of a bronchoscope or other electrosurgical device **118**, as previously described with reference to FIG. **1**. A distal end **105** of the sheath **103** is positioned in the vicinity of the target region **202**. The primary electrode **207** is slidably received within the sheath **103**, with a distal end **209** of the primary electrode **207** at or near the distal end **105** of the sheath **103**. Specifically, FIG. **2**, for example, shows the distal end **209** of the primary electrode **207** positioned just short of the distal end **105** of the sheath **103**. In turn, the secondary electrode **211** is slidably received within the primary electrode **207**, with the distal end **213** of the secondary electrode **211** positioned just within the distal end **209** of the primary electrode **207**.

[0050] Referring to FIG. **3**, the sheath **103**, the primary electrode **207**, and the secondary electrode **211** are positioned once the sheath **103** has been moved closer to the target region **202**. The sheath **103** may be moved toward the target region **202** using a sheath actuator, as described below with reference to FIGS. **6A-7B**. As contrasted with FIG. **2**, in FIG. **3**, the distal end **105** of the sheath **103** has been moved closer to the target region **202**. Because the primary electrode **207** and the secondary electrode **211** have not been separately moved through the manipulation of a user interface (not shown), the primary electrode **207** and the secondary electrode **211** have moved in concert with the sheath **103**, traveling a same distance in a same direction as the sheath **103**. The distal end **209** of the primary electrode **207** remains positioned just short of the distal end **105** of the sheath **103**, and the distal end **213** of the secondary electrode **211** remains positioned just within the distal end **209** of the primary electrode **207**.

[0051] Referring to FIG. **4**, the sheath **103**, the primary electrode **207**, and the secondary electrode **211** are positioned once the primary electrode **207** has been extended from the sheath **103** into the target region **202**. In various embodiments, the secondary electrode **211** moves in concert with the primary electrode **207** as the primary electrode **207** is extended beyond the distal end **105** of the sheath **103**. Thus, the secondary electrode **211** moves in the same direction and moves through the same distance as the primary electrode **207**, as shown in FIG. **4**. The distal end **213** of the secondary electrode **211** remains positioned just within the distal end **209** of the primary electrode **207**.

[0052] Referring to FIG. **5**, the sheath **103**, the primary electrode **207**, and the secondary electrode **211** are positioned once the secondary electrode **211** has been extended from the primary electrode **207**. A distal end **213** of the secondary electrode **211** is deployed at a position across the target region **202** from the primary electrode **207**. In particular embodiments, the secondary electrode **211** is configured as a coilable wire which is constrained within the primary electrode **207** in a straightened form. The secondary electrode **211** may be formed of an alloy, such as nitinol, a nickel-titanium alloy, or other “memory” alloy to regain a certain shape after being released from a confined position. Once the user interface **102** (not shown in FIG. **5**) is manipulated to independently extend the secondary electrode **211** from the primary electrode **207**, a portion of the secondary electrode **211** coils. As a result, the distal end **213** of the secondary electrode **211** augers into tissue at the target region **202**. The augering of the distal end **213** of the secondary electrode **211** may assist in securing the position of the distal end **213** of the secondary electrode **211** during treatment.

[0053] Still referring to FIG. **5**, an insulated section **515** of the secondary electrode **211** stops short of the distal end **213** of the secondary electrode **211**. The insulation **515** electrically insulates the secondary electrode **211** from the primary electrode **207** such that, when electrical current is applied to proximal ends (not shown) of the primary electrode **207** and the secondary electrode **211**, the electrical current may only flow between the distal end **209** of the primary electrode **207** and the uninsulated distal end **213** of the secondary electrode **211**.

[0054] As will be further described below, various embodiments of the user interface **102** facilitate moving the primary electrode **207** and the secondary electrode **211** in concert with the sheath **103**

as the sheath is positioned adjacent the target region **202**, as described with reference to FIG. **3**. Various embodiments of the user interface also facilitate moving the primary electrode **207** and the secondary electrode **211** in concert as they are extended beyond the distal end **105** of the sheath **103**, as described with reference to FIG. **4**. To this end, various embodiments of the user interface **102** may prevent moving the secondary electrode **211** independently of the primary electrode **207** until the primary electrode **207** is extended beyond the distal end **105** of the sheath **103**. Once the primary electrode **207** has been extended, various embodiments of the user interface facilitate moving the secondary electrode **211** independently of the primary electrode **207** to permit separate positioning of the secondary electrode, as described with reference to FIG. **5**. Further, once the primary electrode **207** is deployed at a desirable position, various embodiments of the user interface may prevent the primary electrode **207** from being moved while the secondary electrode **211** is being separately deployed and/or once the secondary electrode **211** has been situated at a desired location. Embodiments of a user interface **102** to coordinate movements of the sheath **103** and electrodes **207** and **211** is explained below with reference to FIGS. **6A-20**.

[0055] Referring to FIGS. **6A** and **6B**, the user interface **102** includes a sheath actuator **604** that is used to position the distal end **105** of the sheath **103**, as previously described with reference to FIG. **3**. The user interface **102** is joined with the electrosurgical apparatus **118** with the coupler **150**, as previously described with reference to FIG. **1**. The electrosurgical apparatus **118**, such as a bronchoscope or another minimally invasive device used for performing diagnostic or therapeutic tasks, conveys the sheath **103** into the body (not shown in FIGS. **6A** and **6B**) near the target region **202**.

[0056] Again referring to FIG. **6A**, the user interface **102** includes a sheath actuator **604** and a sheath lock **606** configured to move the sheath **103** to position the distal end **105** of the sheath **103** at a desired location relative to a target region **202**. In some embodiments, the sheath actuator **604** may be a slidable mechanism incorporating a slidable sleeve **612**. At one end, the slidable sleeve **612** is slidably received within a collar **614** at an end of a housing **610** of the user interface **102**. At an opposing end, the slidable sleeve **612** is joined with the coupler **150**. The slidable sleeve **612** may be locked in position at the collar **614** by the sheath lock **606**. The sheath lock **606** may include a thumbscrew, a spring-loaded locking pin, or another mechanism configured to mechanically engage the slidable sleeve **612** to secure the slidable sleeve **612**—and, in turn, the sheath **103**—in place at a desired location. In some other embodiments, the sheath actuator **604** may, for example, be part of the electrosurgical apparatus **118**. Any such embodiments of the sheath actuator **604** may facilitate movement of the sheath **103**, as further described below.

[0057] Referring to FIG. **6B**, before engaging the sheath actuator **604** to extend the sheath **103**, the sheath **103** and the primary electrode **207** and the secondary electrode **211** received therein are positioned near the target region **202**, as shown in FIG. **2**.

[0058] Referring to FIGS. **7A** and **7B**, manipulation of the sheath actuator **604** illustrates an example of how the sheath **103** may be unlocked and moved into position as previously described with reference to FIG. **3**. In the configuration shown in FIGS. **7A** and **7B**, the sheath actuator **604** has been manipulated to enable the sheath **103** to be moved a distance **719** closer to the target region **202**. Specifically, the sheath lock **606** of the sheath actuator **604** is released to enable movement of the slidable sleeve **612** within the collar **614**. Then, the housing **610** of the user interface **102** is moved a distance **719** relative to the electrosurgical device **118** to move the sheath **103** the same distance **719** toward the target region **702**. Once the distal end **105** of the sheath **103** has reached the desired location relative to the target region **202**, the slidable sleeve **612** may be locked in position at the collar **614** by the sheath lock **606**. In various embodiments of the user interface **102**, the electrodes **207** and **211** move with the housing **610**, so that when the housing **610** is moved to reposition the sheath **103**, the electrodes **207** and **211** move in concert with the sheath **103**. Therefore, as shown in FIG. **7B**, while the distal end **105** of the sheath **103** is advanced toward the target region **202**, the electrodes **207** and **211** move with the sheath **203**. As in FIG. **6B**, the

distal end **209** of the primary electrode **207** remains within the distal end **105** of the sheath **103** and the distal end **213** of the secondary electrode **211** remains within the distal end **209** of the primary electrode **207**.

[0059] Referring to FIG. **8**, in an illustrative sheath actuator **604** and a sheath lock **606** the slidable sleeve **612** is slidably received within the collar **614** of the housing **610**. The slidable sleeve **612** is fixably attached to the coupler **150** that engages the user interface **102** with the electrosurgical apparatus (not shown in FIG. **8**). The sheath lock **606** in the embodiment of FIG. **8** is a thumbscrew that may be loosened to permit movement of the collar **614** fixably attached to the coupler **150** to move the sheath (not shown in FIG. **8**) as previously described with reference to FIGS. **6A-7B**. After the housing **610** has been manipulated to slide the collar **614** relative to the slidable sleeve **612** to move the distal end **105** of the sheath **103** to a desired location, such as described with reference to FIG. **7B**, the sheath lock **606** is reengaged, such as by turning a thumbscrew, to fix the position of the sheath.

[0060] Referring to FIG. **9**, the sheath **103** and the electrodes **207** and **211** extend through the slidable sleeve **612**. As a result, movement of the housing **610**, to which the sheath **103** and the electrodes **207** and **211** are operably coupled, results in movement of the sheath **103** and the electrodes **207** and **211**. A distal end **907** of the sheath lock **606** that extends through the collar **614** mechanically engages the slidable sleeve **612** to control movement of the slidable sleeve **612**. Releasing the sheath lock **606**, such as by loosening a thumb screw, permits the slidable sleeve **612** to be slidably moved relative to the collar **614** by moving the housing **610**, as described with reference to FIG. **7A**. Securing the sheath lock **606**, such as by tightening the thumbscrew, mechanically secures the slidable sleeve **612** in place relative to the collar **614**, preventing further movement of the slidable sleeve **612**, thereby securing the distal end **105** of the sheath **103** in place.

[0061] Referring to FIG. **10**, in various embodiments, the user interface **102** includes control surfaces for positioning the sheath **103** and the electrodes **207** and **211** (none of which are shown in FIG. **10**). The user interface **102** includes the housing **610** that supports components that are moved parallel along an axis **1001** or that are rotated along a curve **1003** around the axis **1001**, as further described below. The user interface **102** includes the sheath actuator **604**, including the collar **614** that receives the slidable sleeve **612** (fully received within the collar **614** and, thus, not shown in FIG. **10**) and the sheath lock **606**. The sheath actuator **604** joins the housing **610** to the coupler **150** which, in turn, couples the user interface **102** with an electrosurgical device (not shown in FIG. **10**). As further described in more detail below, the user interface **102** includes a primary actuator **1010**, which controls movement of the primary electrode **207** (not shown in FIG. **10**), and a secondary actuator **1020**, which controls movement of the secondary electrode **211** (not shown in FIG. **10**).

[0062] The primary actuator **1010** includes a depressible actuator lock **1012** that extends through an actuator opening **1014** in the primary actuator **1010**. The primary actuator **1010** is slidably engaged with the housing **610**. The actuator lock **1012** is hingably or flexibly mounted on the primary actuator **1010**. Depressing the actuator lock **1012** partially moves the actuator lock **1012** through the actuator opening **1014** and a corresponding opening or recess (not shown in FIG. **10**) in the housing **610** to disengage the primary actuator **1010** from the housing **610**. As a result, depressing the actuator lock **1012** permits the primary actuator **1010** to slide along the axis **1001**, as further described below. The secondary actuator **1020** includes an actuator knob **1022** that is engageable to rotate the secondary actuator **1020** through the curve **1003** around the axis **1001**, as also further described below. As also further described below, in various embodiments, actuator interlocks restrict movement of the secondary actuator **1020** until the primary actuator **1010** is moved to extend the primary electrode **207** (not shown in FIG. **10**), and restrict movement of the primary actuator **1010** once the secondary actuator **1020** is moved to extend the secondary electrode **211**.

[0063] Referring to FIG. **11**, various components of the user interface **102**, including portions of the housing **610**, the primary actuator **1010**, and the secondary actuator **1020** illustrate the interrelationship of the components in various embodiments. The housing **610** (FIG. **10**) includes a

first housing section **1131** and a second housing section **1133**. The housing sections **1131** and **1133** have hollow interiors to receive and permit movements of other components arranged therein. A first housing section **1131** internally supports a locking rack **1128** that engages the actuator lock **1012**. More specifically, the locking rack **1128** includes recesses having openings facing inwardly into the housing **610** to permit selective engagement with the actuator lock **1012**. The second housing section **1133** also may include a depth scale **1134** that may be used to visually gauge a position of the primary electrode **207** based on a position of the primary actuator **1010** relative to the housing **610**. A second housing section **1133** threadably supports the sheath lock **606**, which is part of the sheath actuator **604**, as previously described with reference to FIGS. **6A-9**. The housing sections **1131** and **1133** are matable sections, joinable by adhesives or fasteners, such as screws (not shown in FIG. **11**).

[0064] In various embodiments, primary actuator sections **1111** and **1113** are slidably received around the housing section **1131** and **1133**. The primary actuator sections **1111** and **1113** have generally hollow interiors to slidably receive the housing sections **1131** and **1133** therebetween. A first primary actuator section **1111** defines the actuator opening **1014** that receives the actuator lock **1012**. The actuator lock **1012** has a base **1124** that is fixably securable to the first primary actuator section **1111** and around which the actuator lock **1012** partially rotates into an opening or recess (not shown in FIG. **11**) in the housing **610** when the actuator lock **1012** is depressed. At an end opposite the base **1124**, the actuator lock **1012** also supports a pin support **1126** that holds a pin **1127** that engages the locking rack **1128** of the first housing section **1131** when the actuator lock **1012** is not depressed.

[0065] In various embodiments, the actuator lock **1012** is biased into a locking position where the pin support **1126** causes the pin **1127** to engage the locking rack **1128** when the actuator lock **1012** is released. The actuator lock **1012** may be biased by rigidity of the actuator lock **1012** causing the actuator lock **1012** to resume its undeformed position when the actuator lock **1012** is released. Alternatively, the actuator lock **1012** may be spring loaded by a spring actuator (not shown) positioned between the actuator lock **1012** and the housing **610**. The primary actuator sections **1111** and **1113** are joinable by adhesives or fasteners, such as screws (not shown in FIG. **11**).

[0066] Another portion of the primary actuator **1010** is a secondary actuator guide, comprised of guide sections **1151** and **1153** couplable to the primary actuator sections **1111** and **1113**. As described in more detail with reference to FIGS. **22** and **23**, the guide sections **1151** and **1153** are joinable at their ends to form an annular tube and, between their respective edges, define a helical channel that receive guide members **1136** and **1138** extending outwardly from secondary actuator sections **1121** and **1123**. The engagement of the guide members **1136** and **1138** with the helical channel defined by edges of the guide sections **1151** and **1153**, with reference to FIG. **10**, cause the secondary actuator **1020** to advance along the axis **1001** when the secondary actuator **1020** is rotated through a curve **1003** around the axis **1001**.

[0067] In various embodiments, secondary actuator sections **1121** and **1123** are rotatably mounted between the housing sections **1131** and **1133**. The secondary actuator sections **1121** and **1123** are generally hollow to receive therebetween other components of the user interface **102**. As previously described, each of the secondary actuator sections **1121** and **1123** outwardly support the guide members **1136** and **1138** that engage the helical channel defined by edges of the guide sections **1151** and **1153**. Ends **1129** and **1139** of the respective secondary actuator sections **1121** and **1123** are shaped to engage the actuator knob **1022** used to rotate the secondary actuator **1020**, as will be further described below with reference to FIGS. **16A** and **17A**. The secondary actuator sections **1121** and **1123** are joinable by adhesives or fasteners, such as screws (not shown in FIG. **11**).

[0068] In various embodiments, the primary actuator **1010** and the secondary actuator **1020** include actuator interlocks to control relative movement of the actuators **1010** and **1020**. In various embodiments, a first secondary actuator half **1121** may support a recess **1137** and a locking member **1139** to control relative movements of the primary actuator **1010** and the secondary

actuator **1020**. The recess **1137** may be configured to receive the pin support **1126** extending from the actuator lock **1012** to enable the actuator lock **1012** to be depressed to advance the primary actuator **1010**. However, after the primary actuator **1010** is moved, the actuator lock **1012** is released, and the secondary actuator **1020** is rotated, the rotation of the secondary actuator **1020** results in the recess **1137** being displaced from under the pin support **1126**. As a result of the displacement, the actuator lock **1012** is no longer depressible because a body of the secondary actuator **1020** blocks the pin support **1126**, thereby preventing depressing of the actuator lock **1012**. However, after the secondary actuator **1020** is returned to its starting position, the recess **1137** again rotates beneath the pin support **1126**, allowing depressing of the actuator lock **1012** to permit movement of the primary actuator **1010**.

[0069] Similarly, to prevent rotation of the secondary actuator **1020** before the primary actuator **1010** is moved to deploy the primary electrode **207**, the locking member **1139** may engage a notch (not shown) in the housing **610**. After the actuator lock **1012** is depressed and the primary actuator **1010** is moved relative to the housing **610** to deploy the primary electrode **207**, the locking member **1139** clears the housing **610**. It should be noted that the recess **1137** will continue to receive the pin support **1126** as long as the actuator lock **1012** is depressed, continuing to prevent rotation of the secondary actuator **1020**. Once the actuator lock **1012** is disengaged, the secondary actuator **1020** is rotatable to deploy the secondary electrode **211** and to block the actuator lock **1012** from being engaged to permit movement of the primary actuator **1010**. Thus, in sum, the actuator interlocks ensure that the primary actuator **1010** be moved to deploy the primary electrode **207** before the secondary actuator **1020** may be rotated. Then, once the primary actuator **1010** has been moved to deploy the primary electrode **207** and the secondary actuator **1020** is rotated from its starting position, the actuator interlocks prevent the primary actuator **1010** and the primary electrode **207** from being moved until the secondary actuator **1020** is moved to retract the secondary electrode **211** to its original position.

[0070] The user interface **102** also includes a sheath mount **1135** that is receivable between the housing sections **1131** and **1133** to mechanically engage the housing **610** with the sheath **103**. As a result, as described with reference to FIGS. **6A-9**, movement of the housing **610** extends or retracts the sheath **103**. The user interface also includes electrode sliders coupled with the respective electrodes **207** and **211**. A primary electrode slider **1145** is mechanically engageable by the primary actuator sections **1113** and **1133** so that sliding the primary actuator **1010** advances or retracts the primary electrode slider **1145** to advance or retract the primary electrode **207**, respectively. A secondary electrode slider (not shown in FIG. **11**) mechanically engageable by the secondary actuator sections **1121** and **1123** is slidably received within the primary electrode slider **1145**. Because the secondary actuator sections **1121** and **1123** are rotatably moved, as further described below, the secondary electrode slider is also rotatably received between the secondary actuator sections **1121** and **1123**.

[0071] A flexible wiring harness **1150** is configured to receive one or more conductors of the electrical conductor **130** (FIG. **1**) at a port on the housing **610** (not shown in FIG. **11**), and to electrically connect with flexible leads **1152** and **1154**, each of which connects with one of the electrodes **207** and **211**. The flexible leads **1152** and **1154** are configured to remain electrically connected with the electrodes **207** and **211** as proximal ends of the electrodes **207** and **211** are moved within the user interface **102**.

[0072] Additionally, a flexible fluid coupling **1160** extends from a fluid port (not shown in FIG. **11**) on the housing **610** to an interior of the primary electrode slider **1145** to convey fluid into a lumen defined within the primary electrode **207**. The fluid port receives the tube **132** from the infusion pump **116** (FIG. **1**) at the housing **610** to receive a flow of conductive fluid. The flexible fluid coupling **1160** may be coiled within the housing **610** to permit extension and contraction of the fluid coupling **1160** with the movement of the primary electrode slider **1145** relative to the housing **610**.

[0073] As further described below with reference to FIGS. 27 and 28, the coupler 150 includes a slidable locking body 1180 that is slidably received between a slidable mount 1182 and a retaining ring 1184. The slidable mount 1182 is coupled with the housing 610. As further described below, once the housing 610 is positioned to engage the electrosurgical device 118 (not shown in FIG. 11), the locking body 1180 is slid into place to secure the connection, as further described with reference to FIGS. 27 and 28.

[0074] Referring to FIGS. 12A-21B, operation of the user interface 102 and corresponding movements of the sheath 103, the primary electrode 207, and the secondary electrode 211 are described.

[0075] Referring to FIGS. 12A and 12B, the distal end 105 of the sheath 103 is positioned adjacent to the target region 202. As previously described with reference to FIGS. 6A-7B, in various embodiments, the sheath actuator 604 enables the sheath 103 to be positioned by releasing the sheath lock 606 and moving the housing 610. For example, referring again to FIGS. 6A-7B, a position of the sheath 103 is controlled by sliding the slidable sleeve 612 within the collar 614, then securing the sheath 103 at the desired location by reengaging the sheath lock 606. When the distal end 105 of the sheath 103 is deployed adjacent to the target region 202, a distal end 209 of the primary electrode 207 lies just within the distal end 105 of the sheath 103. At the same time, the distal end 213 of the secondary electrode 211 lies just within the distal end 209 of the primary electrode 207. With the distal end 105 of the sheath 103 positioned adjacent the target region 202, the user interface 102 may be used to move the electrodes 207 and 211 to desired positions.

[0076] Referring to FIGS. 13A and 13B, according to various embodiments, positioning the electrodes 207 and 211 begins with depressing the actuator lock 1012 to enable movement of the primary actuator 1010. Depressing the actuator lock 1012 to move the actuator lock 1012 in a direction 1301 disengages the primary actuator 1010 from the housing 610. Specifically, depressing the actuator lock 1012, which is hingably or rotatably coupled with the primary actuator 1010 at the base 1124, causes the pin support 1126 to move the pin 1127 from inward-facing recesses of the locking rack 1128 on the housing 610. With the pin 1127 removed from the locking rack 1128, the primary actuator 1010 is movable relative to the housing 610 to move the primary electrode 207, as described with reference to FIGS. 14A and 14B.

[0077] As previously described, the secondary actuator 1020 is rotatably engaged with the primary actuator 1010. Accordingly, the secondary actuator 1020 remains engaged with the primary actuator 1010 even when the actuator lock 1012 is released to release the primary actuator 1010 from the housing 610. Therefore, depressing the actuator lock 1012 frees the primary actuator 1010 and the secondary actuator 1020 to move collectively, thus enabling the primary electrode 207 and the secondary electrode 211 to be moved collectively.

[0078] Referring to FIGS. 14A and 14B, while a user continues to depress the actuator lock 1012 in the direction 1301, the primary actuator 1010 is moved in a direction 1401. Because the secondary actuator 1020 remains (rotatably) engaged with the primary actuator 1010 as previously described, the primary actuator 1010 and the secondary actuator move collectively the same distance in the direction 1401, as represented in FIG. 14A.

[0079] As a result of the collective movement of the primary actuator 1010 and the secondary actuator 1020, the primary electrode 207 and the secondary electrode 211 move collectively as well. Thus, as depicted in FIG. 14B, the distal end 209 of the primary electrode 207 and the distal end 213 of the secondary electrode 211 move collectively beyond the distal end 105 of the sheath 103 into the target region 202. Thus, by virtue of the engagement of the secondary actuator 1020 with the primary actuator 1010, depressing the actuator lock 1012 and moving the primary actuator 1010 moves both electrodes 207 and 211 collectively.

[0080] As previously described with reference to FIG. 11, with the actuator lock 1012 depressed, in various embodiments, the pin support 1026 on the actuator lock 1012 engages the secondary actuator 1020, preventing the secondary actuator 1020 from being rotated until the actuator lock

1012 is disengaged. As also previously described, the secondary actuator **1020** may include the locking member **1139** that abuts the housing **610**. This arrangement prevents the secondary actuator **1020** from being rotated before the actuator lock **1012** is depressed and the primary actuator **1010** and the secondary actuator **1020** are advanced.

[0081] Referring to FIGS. **15A** and **15B**, once the distal ends **209** and **213** of the primary electrode **207** and the secondary electrode **211**, respectively, have been advanced into the target region **202**, the actuator lock **1012** is released. Because the actuator lock **1012** is biased by its rigidity or by a spring, as described with reference to FIG. **11**, releasing the actuator lock **1012** results in the actuator lock **1012** moving in a direction **1501**. The movement of the actuator lock **1012** causes the primary actuator **1010**—and the rotatably engaged secondary actuator **1020**—to again be engaged with the housing **610**, holding the electrodes **207** and **211** in place. As described with reference to FIG. **11**, when the actuator lock **1012** is released, a pin **1127** mounted in the pin support **1026** moves into recesses in the locking rack **1128** mounted on the housing **610**. Thus, the engagement of the pin **1127** with the locking rack **1128** prevents further movement of the primary actuator **1010** until the actuator lock **1012** is further engaged by a user. Accordingly, with the user releasing the actuator lock **1012**, the distal ends **209** and **213** of the primary electrode **207** and the secondary electrode **211** are secured in the locations to which they were moved as described with reference to FIGS. **14A** and **14B**.

[0082] Referring to FIGS. **16A** and **16B**, with the primary actuator **1010** held in place by the user's release of the actuator lock **1012**, the secondary actuator **1020** is rotated to move the secondary electrode **211** independently of the primary electrode **207**. As shown in FIG. **16A**, the secondary actuator **1020** is moved by a user rotating the actuator knob **1022** in a direction **1601**. As previously described with reference to FIG. **11**, the secondary actuator **1020** supports guide members **1136** and **1138** that are received within the helical channel defined between edges of the guide sections **1151** and **1153**. With the secondary actuator **1020** engaged with the helical channel defined by the guide sections **1151** and **1153**, rotation of the actuator knob **1022** results in helical movement of the secondary actuator **1020**. The rotation of the secondary actuator **1020** thus causes the secondary actuator **1020** to advance in a direction **1602** relative to the primary actuator **1010** and the housing **610**.

[0083] Referring to FIG. **16B**, movement of the secondary actuator **1020** results in the distal end **213** of the secondary electrode **211** extending beyond the distal end **207** of the primary electrode **209**. As previously described with reference to FIG. **5**, the distal end **213** of the secondary electrode **211** may be preformed into a coiled shape, thereby resulting in the secondary electrode **211** forming a coiled shape once the secondary electrode **211** is no longer constrained within the lumen of the primary electrode **207**. In various embodiments, the coiled shape at the distal end **213** of the secondary electrode **211** augers into the tissue of the target region **202**, which secures the secondary electrode **211**—and the primary electrode **207** through which it extends—in position at the target region **202**. The insulated section **515** of the secondary electrode **211** electrically insulates the secondary electrode **211** from the primary electrode **207** except as between their respective distal ends **213** and **209**. With the distal ends **213** and **209** of the electrodes **211** and **207** deployed, a supply of conductive fluid and/or electrical current may be applied to the target region **202** as previously described to effect treatment.

[0084] The actuator interlocks presented by the configuration of the actuators **1010** and **1020** prevent the user from moving the primary actuator **1010** once the secondary actuator **1020** is rotated from its original position. As previously described with reference to FIG. **11**, rotating the secondary actuator **1020** blocks the pin support **1126** of the actuator lock **1012**, thereby preventing a user from depressing the actuator lock **1012** to release the primary actuator **1010** from its engagement with the housing **610** via the pin **1127** and the locking rack **1128**. Thus, the distal end **209** of the primary electrode **207** remains in place as inserted into the target region **202** while the secondary actuator **1020** is moved to extend the distal end **213** of the secondary electrode **211** into

the target region **202**.

[0085] Deployment of the sheath **103** and the electrodes **207** and **211** to permit the application of treatment is described with reference to FIGS. **6A-7B** and **12A-16B**. Conversely, to withdraw and move the electrodes **207** and **211** from the target region **202**, manipulations and the sequence of manipulations of the user interface **102** is reversed, as described with reference to FIGS. **17A-21B**. [0086] Referring to FIGS. **17A** and **17B**, the distal end **213** of the secondary electrode **211** is retracted into the primary electrode **207** by a user rotating the actuator knob **1022** in a direction **1701**. The direction **1701** in which the actuator knob **1022** is rotated to retract the distal end **213** of the secondary electrode **211** from the target region **202** is opposite to the direction **1601** in which the actuator knob **1022** was rotated to extend the distal end **213** of the secondary electrode **211**. Rotation of the actuator knob **1022** results in an opposite, helical movement of the secondary actuator **1020**, resulting the secondary actuator **1020** translating in a direction **1702** relative to the primary actuator **1010** and the housing **610**. The movement of the secondary actuator **1020** withdraws the secondary electrode **211** until the distal end **213** of the secondary electrode **211** again is received within the distal end **209** of the primary electrode **207**. With the secondary actuator **1020** moved to its original position relative to the primary actuator **1010**, the actuator lock **1012** now may be released, as described with reference to FIG. **18A**. It will be appreciated that retraction of the secondary electrode **211** is accomplished by rotating the secondary actuator **1020** while the primary actuator **1010** remains stationary.

[0087] Referring to FIGS. **18A** and **18B**, to prepare for retraction of the primary electrode **207** from the target region **202**, the actuator lock **1012** is depressed by a user in a direction **1801**. Depressing the actuator lock **1012** does not result in any movement of the distal ends **209** and **213** of the electrodes **207** and **211**, respectively, just as engagement of the actuator lock **1012** did not result in movement of the electrodes **207** and **211** when the actuator lock **1012** was depressed and released as previously described with reference to FIGS. **13A** and **13B** and FIGS. **15A** and **15B**, respectively.

[0088] Referring to FIGS. **19A** and **19B**, with the actuator lock **1012** depressed, the primary actuator **1010** is moved in a direction **1901** to withdraw the distal end **209** of the primary electrode **207** from the target region **202**. As previously described with reference to FIGS. **14A** and **14B**, because the secondary actuator **1020** remains rotationally engaged with the primary actuator **1010**, the secondary actuator **1020** also moves a same distance and in the same direction **1901** as the primary actuator **1010**. As a result, the distal ends **209** and **213** of the electrodes **207** and **211** are moved collectively and withdrawn from the target region **202**. After the primary actuator **1010** is fully retracted in the direction **1901**, the distal end **209** of the primary electrode **207** is received within the distal end **105** of the sheath. Further, because the secondary actuator **1020**—and, thus, the secondary electrode **211**—moves in concert with the primary actuator **1010**, the distal end **213** of the secondary electrode **211** remains within the distal end **209** of the primary electrode **207** as the distal end **209** of the primary electrode **207** is withdrawn within the distal end **105** of the sheath **103**.

[0089] Referring to FIGS. **20A** and **20B**, once the distal ends **209** and **213** of the electrodes **207** and **211**, respectively, are withdrawn within the distal end **105** of the sheath **103**, the actuator lock **1012** is released. Upon release of the actuator lock **1012**, the actuator lock **1012** moves in a direction **2001**. As a result, the pin **1127** held by the pin support **1126** reengages the locking rack **1128** to hold the primary actuator **1010** in place. Further, as previously described, the actuator interlocks that prevent the secondary actuator **1020** from being rotated, such as by the locking member **1139** extending from the secondary actuator **1020** engaging the housing **610**, prevents rotation of the secondary actuator **1020** while the actuators **1010** and **1020** have resumed a starting position as described with reference to FIGS. **12A** and **12B**.

[0090] Referring to FIGS. **21A** and **21B**, with distal ends **209** and **211** of the electrodes **207** and **211**, respectively, withdrawn within the distal end **105** of the sheath **103**, the sheath **103** itself may

be withdrawn. In an operation opposite that depicted in FIGS. 7A and 7B, the sheath lock **606** is released and the housing **610** is moved along the slidable sleeve **612** in a direction **2101** away from the coupling **150**. Because the primary actuator **1010** is locked to the housing by the actuator lock **1012**, and the secondary actuator **1020** is rotatably secured to the primary actuator **1010**, the primary actuator **1010** and the secondary actuator **1020** move in concert with the housing **610** in the direction **2101**. The sheath **103** and the insertion tube **119** (FIG. 1) of the electrosurgical device **118** may then be withdrawn from the body. Alternatively, without withdrawing the sheath as described with reference to FIGS. 21A and 21B, once the electrodes **207** and **211** are withdrawn into the sheath as described with reference to FIGS. 19A-20B, the sheath **103** may be withdrawn from the body without first withdrawing the sheath **103** by engaging the sheath lock **606**.

[0091] As previously described with reference to FIGS. 11, 16A, and 17A, the secondary actuator **1020** supports guide members **1136** and **1138** that engage a helical channel defined by edges of guide sections **1151** and **1153**. Referring to FIG. 22, the guide sections **1151** and **1153** are mated together into a guide sleeve **2202** as they are when joined with the primary actuator **1010**. The guide sections **1151** and **1153** may be joined at ends **2215** and **2217**. Specifically, as shown in FIG. 23, sockets **2330** may be supported by the guide sections **1151** and **1153** enabling the guide sections to be connected by screws, dowels, or other fasteners.

[0092] Between the ends **2215** and **2217** of the guide sleeve **2202**, edges **2211** and **2213** of the guide sections **1151** and **1153** define a helical channel **2201**. The helical channel **2201** guides the movement of the support members **1136** and **1138** to cause the secondary actuator **1020** to translate in response to rotation of the secondary actuator as described with reference to FIGS. 16A and 17A.

[0093] In various embodiments, the generally helical channel **2201** has a varied pitch between the ends **2215** and **2217** of the guide sleeve **2202**. In various embodiments, the pitch may vary from a rearward end **2215**, where the secondary actuator **1020** begins its helical movement to extend the secondary electrode **207**, toward a forward end **2217**. More specifically, in various embodiments, the pitch of the helical channel is varied to reduce a distance of travel of the secondary actuator **1020** along the axis **1001** of the user interface **102** (not shown in FIG. 22) per unit of rotation of the secondary actuator **1020** from the rearward end **2215** toward the forward end **2217**.

[0094] In various embodiments, the pitch is varied in this manner to reduce the rotational force to be applied by a user in turning the actuator knob **1022** to motivate the secondary actuator **1020**. For example, considering FIGS. 5 and 16B, as the distal end **213** of the secondary electrode **211** is advanced into the target region **202**, the distal end **213** of the secondary electrode **211** may encounter increased resistance. Part of this resistance results from the distal end **213** of the secondary electrode **211** frictionally engaging a mass in the target region along an increasing length of the secondary electrode **211** as a longer section of the secondary electrode **211** is extended further beyond the distal end **209** of the primary electrode **207**. Part of this resistance may also result from the curvature of the coil at the distal end **213** of the secondary electrode **211** encountering an increasing degree of resistance in augering into the mass at the target region **202**. Correspondingly, greater force may be involved at the start of withdrawal of the secondary electrode **211** in frictionally engaging a greater mass of tissue than when the secondary electrode **211** is closer to being fully retracted into the distal end **209** of the primary electrode **207**. Further, when a portion of the secondary electrode **211** near the distal end **213** is formed into a coiled shape using a memory alloy, withdrawing the secondary electrode **207** may involve application of additional force in seeking to draw the secondary electrode into a deformed, straightened shape that the secondary electrode **211** assumes when confined within the primary electrode **207**.

[0095] As a result, in deploying the secondary electrode **211**, more force may be involved in extending the secondary electrode **211** as the secondary electrode **211** extends further beyond the distal end **209** of the primary electrode **207** into the target region **202**. As a result, a greater degree of rotational force may be involved in rotating the actuator knob **1022** of the secondary actuator **1020** as the secondary actuator **1020** moves toward the forward end **2217** of the guide sleeve **2202**.

Correspondingly, more force may be involved the initial portion of withdrawing the secondary electrode **211** than when the secondary electrode **211** has been or nearly has been fully retracted into the primary electrode **207**. Therefore, a greater degree of rotational force may be involved in rotating the actuator knob **1022** of the secondary actuator **1020** as the secondary actuator **1020** first moves away from the forward end **2217** of the guide sleeve **2202**.

[0096] According to various embodiments, the pitch of the helical channel **2201** may be varied between the trailing end **2215** and the forward end **2217** of the guide sleeve **2202**. Specifically, the pitch of the helical channel **2201** may be varied to reduce a distance of travel of the second actuator **1020** along the axis **1001** per unit of rotation through the curve **1003** around the axis **1001** toward the forward end **2217** of the guide sleeve **2202** facing a forward end of the user interface **2202**. By reducing the distance of travel of the second actuator **1020** toward the forward end of the guide sleeve **2202**, the increased force along the axis **1001** is effectively spread over a greater degree of rotation of the second actuator **1020**. Thus, while lateral resistance to moving the secondary electrode **211** along the axis **1001** may increase at a forward end **2217** of the guide sleeve **2202**, the force involved in rotating the actuator knob **1022** to rotate the secondary actuator **1020** does not increase as much.

[0097] Referring to FIG. **23**, because the pitch of the helical channel **2201** is defined by the edges **2211** and **2213** of the guide sections **1151** and **1153**, respectively, a pitch of the edges **2211** and **2213** is varied to define a helical channel **2201** of a desired shape. For example, considering the first guide section **1153**, at a first point **2301** toward a rearward end **2345** of the first guide section **1153**, a pitch angle α of the edge **2213** (as measured tangentially to the edge **2213** relative to the axis **1001**) is greater than a pitch angle β at a second point **2302** moving toward the forward end **2347** of the first guide section **1153**. Similarly, the pitch angle β at the second point **2302** is greater than a pitch angle γ at a third point **2303** moving further toward the forward end **2347** of the guide section **1153**. A corresponding arrangement is repeated with the second guide section **1151**, with a pitch angle along the edge **2211** becoming less moving from a rearward end **2341** of the second guide section **1151** toward the forward end **2343**. As a result, despite increased resistance along the axis **1001**, rotational resistance applied to the secondary actuator **1020** is reduced by the decreasing pitch of the helical channel **2201** (FIG. **22**) defined by the decreasing pitch of the edges **2211** and **2213** of the respective guide sections **1151** and **1153**.

[0098] In addition to varying the pitch of the helical channel **2201** to facilitate deployment and withdrawal of the secondary electrode **207**, a cross-section of the wire used as the secondary electrode **207** also may ease the deployment and withdrawal of the secondary electrode **207**. Referring to FIGS. **24-26**, the secondary electrode **207** may include a wire having portions **2410** and **2420** of different thicknesses along its length.

[0099] Referring to FIG. **24**, a first portion **2410** of the secondary electrode **207** may have a circular cross-section with a first thickness **2412**. A second portion **2420** leading to the distal end **213** of the secondary electrode **211** may have a flat or rectangular cross-section having a second thickness **2422** that is less than the first thickness **2412**. In an illustrative embodiment, the first thickness **2412** of the circular cross-section of the first portion **2410** may be 0.015 inches, and the second thickness **2422** of the second portion may be 0.009 inches. In such a configuration, a theoretical moment of inertia for the first portion **2410** is more than twice that of a theoretical moment of inertia for the second portion **2420**. The greater theoretical moment of inertia of the first portion **2410** thus should improve the force transmission of the first portion **2410** in advancing the secondary electrode **207** without impeding the capacity of the second portion **2420** to assume its coiled configuration upon deployment. The first thickness **2412** is aligned with an axis **2430** that defines a plane in which the second portion **2402** will coil, as depicted in FIG. **25**.

[0100] Referring to FIG. **26**, the secondary electrode **211** is in an uncoiled configuration. The second portion **2420** may have a second width **2624** that is wider than the second thickness **2422** of the second portion **2420** and wider than the first thickness **2412** of the first portion **2410**. In a non-

limiting example, the first thickness **2412** may be 0.015 inches, the second thickness may be 0.009 inches, and the second width may be 0.020 inches.

[0101] A secondary electrode **211** with the first portion **2410** having a circular cross-section provides good column strength and force transmission for motivating the secondary electrode **211** along its length. The column strength and force transmission are helpful in driving the secondary electrode **211** through the lumen within the primary electrode **207** and in extending the secondary electrode **211** into a tissue at a target region, as depicted in FIG. 5. By contrast, with the second portion **2420** having a reduced thickness in the plane in which the second portion **2420** of the secondary electrode **211** is to coil makes it easier for the secondary portion to assume its coiled shape. Using the exemplary dimensions, the moment of inertia of the second portion **2420** is less than half of that of the first portion **2410**, reducing the force required to coil and uncoil the second portion **2410**. Having a second width **2624** that is larger than the second thickness **2422** and larger than the first thickness **2412** improves the column strength and force transmission of the second portion **2420** to keep the second portion **2420** from buckling, while still having a thinner second thickness **2422** that facilitates the coiling of the second portion **2420**.

[0102] Referring to FIG. 27, the coupler used to secure the user interface **102** with the electrosurgical device **118** includes a slidable mounting mechanism **2710** and a locking body **2720**. In various embodiments, the slidable mounting mechanism **2710** is secured to the slidable sleeve **612** extending from the housing of the user interface (not shown in FIG. 27) and fits around the slidable sleeve **612**. The slidable sleeve **612** has an internal width **2791** that is sized to receive a flange **2754** at an end of a device interface **2752**. The flange **2754** has an outer width **2793** that is less than the internal width **2791** of the slidable sleeve **612** so that the flange **2754** is receivable within an end of the slidable sleeve **612**. The device interface **2752** has an outer width **2795** that is less than the outer width **2793** of the flange **2754** that it abuts. The outer width **2795** of the device interface **2795** and the outer width **2793** of the flange **2754** are considered in the configuration of the locking body **2720**, as further described below with reference to FIG. 28.

[0103] The slidable mounting mechanism **2710** includes a base portion **2712** that is fixed, fixable, or connected to the slidable sleeve **612** (the slidable mounting mechanism **2710** is shown in FIG. 27 prior to being fixably connected to the slidable sleeve **612**). The slidable mounting mechanism **2710** also includes one or more projections **2714** that are configured to receive retaining clips **2734** extending from a retaining ring **2730** to secure the lock plate **2720** to the slidable mounting mechanism **2710**, as further described below.

[0104] The slidable mounting mechanism **2710** also supports a locking pin **2716**. In various embodiments, the locking pin **2716** is spring-loaded or otherwise biased to extend outwardly from the slidable mounting mechanism **2710** to engage a locking slot in the locking body **2720** to prevent the locking body **2720** from sliding. The locking pin **2716** may be manually retracted away from the locking body **2720** to permit the locking body **2720** to be moved to an unlocked position.

[0105] In various embodiments, the slidable mounting mechanism **2710** includes a torque transfer mechanism to transfer torque between the electrosurgical instrument or device **118** (FIG. 1) and the user interface **102**. In various embodiments, the torque transfer mechanism includes a linkage **2728** that is received within a channel **2718** when the locking body **2720** is in a locked position. The linkage **2728** and the channel **2718** thus transfer torque between the locking body **2720** that is engaged with the electrosurgical instrument or device **118** and the slidable sleeve **612**. The linkage **2718** and channel **2728** thus absorb and/or transfer torque between the electrosurgical instrument or device **118** and the slidable sleeve **612**, rather than, for example, the torque being exerted on the retaining ring **2730** and/or the locking pin **2716**. In various embodiments, torque also may be absorbed and transferred by strengthening the locking pin **2716** and/or tightening and strengthening the mounting of the retaining ring **2730** to the lock plate **2720**.

[0106] The locking body **2720** has a base plate **2722** configured to slide across the slidable mounting mechanism **2710** and to hold the flange **2754** in place within the slidable sleeve **612** to

secure the user interface **102** to the electrosurgical device **118**. As further described with reference to FIG. **28**, the base plate **2722** defines an opening having differently-sized sections that alternately permit insertion of the flange **2754** into the slidable sleeve **612** and prevent removal of the flange **2754** from the slidable sleeve **612**. The locking body **2720** supports a hood **2724** that extends over the combination formed by the surgical device interface **2752** with the user interface **102** via the slidable sleeve **612**. As previously mentioned, the locking body **2720** also supports the second indicator tab **2728**. The second indicator tab **2728** aligns with the first indicator tab **2718** on the slidable mounting mechanism **2710** when the locking body **2720** is in a locked position to provide visual confirmation when the locking body **2720** is in a locked position.

[0107] The locking body **2720** is slidably secured to the slidable mounting mechanism with a retainer ring **2730**. The retainer ring **2730** includes a ring **2732** having an inner diameter **2799** that is sized to receive the flange **2754** extending from the surgical device interface **2752** therethrough. Extending from the ring **2732** are one or more retaining clips **2734**. The retaining clips **2734** are sized to fit through slots in the base plate **2722** of the locking body, as further described below with reference to FIG. **28**. Once the retaining clips **2734** are extended through the slots in the base plate **2722** of the locking body **2720**, the retaining clips are secured onto and/or around the projections **2714** on the slidable mounting mechanism **2710**. Once the retaining clips **2734** are extended through the slots on the locking body **2720** and secured onto the projections **2714** on the slidable locking mechanism **2710**, the locking body **2720** is slidably constrained to move across the slidable mounting mechanism **2710** to lock and unlock the user interface **102** with the electrosurgical device **118**.

[0108] Referring to FIG. **28**, the base plate **2722** defines two retaining slots **2895** through which the retaining clips **2734** (FIG. **27**) extend from the retaining ring **2730**. The retaining slots **2895** are sized to slidably receive the retaining clips **2734** so that the locking body **2720** can slide in a first direction **2815** or a second direction **2817** across the retaining clips **2734**. The ring **2732** of the retaining ring **2730** lies across the base plate **2722** to hold the locking body **2720** to the slidable mounting mechanism **2710** (FIG. **27**). The locking body **2720** also supports at least one socket **2820** to receive the locking pin **2716** (FIG. **27**) extending from the slidable locking mechanism **2710**. The socket **2820** is positioned to engage the locking pin **2716** when the locking body **2720** is slid into a locked position over the surgical device interface **2752**.

[0109] The hood **2724** extends from the locking plate **2722** to cover the connection between the surgical device interface **2752** and the user interface **102**. To allow the locking body **2720** to move in the second direction **2817** without the hood **2724** being blocked by a body of the electrosurgical device **118** (FIG. **1**), a lower edge **2825** of the hood **2724** is shaped to define a recess **2827**. When the locking body **2720** is moved in the second direction **2817** to move the locking body **2720** into a locked position, the recess **2827** receives the body of the electrosurgical device **118**.

[0110] The base plate **2722** of the locking body **2720** defines an opening **2810** through which, as described with reference to FIG. **27**, the flange **2754** on the surgical device interface **2772** may be inserted into the slidable sleeve **612**. More specifically, a first section **2801** of the opening has a first width **2811** and a conjoined second section **2803** with a second width **2813**. The first width **2811** of the first section is large enough to receive the outer width **2793** of the flange **2754** therethrough, while the second width **2813** of the second section **2803** is large enough to receive the width **2795** of the surgical device interface **2752** but not to allow the outer width **2793** of the flange **2754** to pass therethrough.

[0111] To lock the user interface **102** with the electrosurgical device **118** (not shown in FIG. **28**), the locking body **2720** is slid in the first direction **2815** to position the first section **2801** over the opening in the base portion **2712** of the slidable mounting mechanism **2710** that leads into the slidable sleeve **612** (not shown in FIG. **28**). The flange **2754** that extends from the surgical device interface **2752** is then inserted through the first section **2801** and into the slidable sleeve **612**. To secure the surgical device interface **2752** in place, the locking body is slid in the second direction

2817. As a result, the second section **2803** is moved over the opening in the base portion **2712** of the slidable mounting mechanism **2710**, and an edge of the locking body **2722** slides over and abuts the flange **2754**. In this locked position, the edge of the locking plate **2722** around the second section **2803** cover the flange **2754** and holds the flange **2754** in place. Also, with the locking body **2720** in this locked position, the locking pin **2716** extends into the socket **2820**. The locking pin **2716** blocks movement of the locking plate in the first direction **2815** until the locking pin **2716** is withdrawn from the socket **2820**.

[0112] To uncouple the user interface **102** from the electrosurgical device **118**, a user engages the locking pin **2716** to slide it out of the socket **2820** to permit sliding movement of the locking body **2720**. With the locking pin **2716** withdrawn, the locking body **2720** is slid in the first direction **2815** so that the base plate **2722** moves away from the surgical device interface **2752** with the first section **2801** of the opening **2810** over the flange **2752**. The flange **2754** of the surgical device interface **2752** can now be withdrawn through the locking body **2720**, ending the connection between the surgical device interface **2752** and the slidable sleeve **612** of the user interface **102**.

[0113] Referring to FIG. **29**, an illustrative method **2900** of positioning electrodes for treatment is provided. The method **2900** starts at a block **2905**. At a block **2910**, a distal end of a sheath that contains a primary electrode and a secondary electrode is moved adjacent to a target region, as described with reference to FIGS. **6A-7B**. At a block **2920**, a primary actuator, that is operably coupled with the primary electrode and a secondary actuator that is operably coupled to the secondary electrode and that is movably engaged with the primary actuator, is slid to a first position to motivate distal ends of the primary electrode and the secondary electrode relative to the target region, as described with reference to FIGS. **14A** and **14B**. At a block **2930**, the secondary actuator is rotated relative to the primary actuator to cause the secondary actuator to travel independently of the primary actuator along a helical path to a second position to motivate the distal end of the secondary electrode to move independently of the primary electrode relative to the target region, as previously described with reference to FIGS. **16A** and **16B**. The method **2900** ends at a block **2935**, with the electrodes now positioned.

[0114] Referring to FIG. **30**, an illustrative method **3000** of motivating an implement through a helical path having a varied pitch is provided. The method **3000** starts at a block **3005**. At a block **3010**, an elongated implement is coupled at a proximal end thereof to an actuator that is movable along an axis, as described with reference to FIG. **11**. At a block **3020**, the implement is motivated by rotatably moving the actuator through a generally helical path around the axis, the helical path having a pitch that is varied to change a distance traveled by the actuator along the axis per unit of rotation of the actuator, as described with reference to FIGS. **16A**, **16B**, **22**, and **23**. The method **3000** ends at a block **3025**, with the actuator having moved the implement.

[0115] Referring to FIG. **31**, an illustrative method **3100** of securing devices together is provided. The method **3100** starts at a block **3105**. At a block **3110**, a locking body is positioned into an open position, where the locking body defines an opening with a first section having a first width and a second section having a second width that is smaller than the first width. The locking body is slidably mounted on one of a first device that supports a first coupling and a second device that supports a second coupling. The first section is disposed between the first coupling and the second coupling when the locking body is positioned into the open position, as described with reference to FIG. **28**. At a block **3120**, a connection is formed by inserting the first coupling within the second coupling where one of the first and second couplings supports a flange having a flange width that is smaller than the first width and larger than the second width, as described with reference to FIG. **28**. At a block **3130**, the locking body is repositioned into a closed position in which an edge of the locking body around the second section abuts the flange so that the coupling that supports the flange is prevented from being withdrawn from the connection, as previously described with reference to FIG. **28**. The method **3100** ends at a block **3135**, with the couplings secured together by the locking body.

[0116] It will be appreciated that the detailed description set forth above is merely illustrative in nature and variations that do not depart from the gist and/or spirit of the claimed subject matter are intended to be within the scope of the claims. Such variations are not to be regarded as a departure from the spirit and scope of the claimed subject matter.

Claims

1. (canceled)
2. A method comprising: moving a distal end of a sheath that contains a primary electrode and a secondary electrode adjacent to a target region; sliding a primary actuator that is operably coupled with the primary electrode and a secondary actuator that is operably coupled to the secondary electrode and that is movably engaged with the primary actuator to a first position to motivate distal ends of the primary electrode and the secondary electrode relative to the target region; and rotating the secondary actuator relative to the primary actuator to cause the secondary actuator to travel independently of the primary actuator along a helical path to a second position to motivate the distal end of the secondary electrode to move independently of the primary electrode relative to the target region.
3. The method of claim 2, further comprising: preventing the secondary actuator from being rotated relative to the primary actuator until the primary actuator is moved to the first position; and preventing the primary actuator from sliding after the secondary actuator is rotated to move the secondary actuator to the second position.
4. The method of claim 2, further comprising activating a primary actuator lock to selectively disengage the primary actuator from a housing to enable slidable movement of the primary actuator relative to the housing.
5. The method of claim 4, wherein activating the primary actuator lock includes depressing a release mechanism configured to permit the primary actuator to become disengaged from the housing while the release mechanism is depressed and further configured to permit the primary actuator to engage the housing when the release mechanism is released.
6. The method of claim 2, wherein rotating the secondary actuator includes at least one guide member within the secondary actuator engaging with one or more guide sections forming the helical path.
7. The method of claim 2, wherein moving the distal end of the sheath includes releasing a sheath lock and moving a housing portion of a user interface.
8. The method of claim 7, wherein moving the distal end of the sheath includes reengaging the sheath lock once the distal end is adjacent the target region.
9. The method of claim 2, wherein sliding the primary actuator includes depressing an actuator lock to disengage the primary actuator from a housing of a user interface.
10. The method of claim 9, wherein sliding the primary actuator further includes releasing the actuator lock in response to advancing the primary electrode and the secondary electrode into the target region.
11. The method of claim 2, further comprising supplying a conductive fluid to the target region.
12. The method of claim 2, further comprising supplying an electrical current across the primary electrode and the secondary electrode.
13. A method comprising: deploying an electrosurgical instrument including a primary electrode and a secondary electrode controlled through a user interface, the user interface including a housing, a primary actuator coupled to the primary electrode, and a secondary actuator coupled to the secondary electrode; moving a distal end of a sheath that contains the primary electrode and the secondary electrode adjacent to a target region using a housing portion of the user interface; manipulating the primary actuator to concurrently move distal ends of the primary electrode and the secondary electrode to a first position relative to the target region; and rotating the secondary

actuator relative to the primary actuator to cause the secondary actuator to travel independently of the primary actuator along a helical path to a second position to motivate the distal end of the secondary electrode to move independently of the primary electrode relative to the target region.

14. The method of claim 13, further comprising activating a primary actuator lock to selectively disengage the primary actuator from the housing to enable slidable movement of the primary actuator relative to the housing.

15. The method of claim 14, wherein activating the primary actuator lock includes depressing a release mechanism configured to permit the primary actuator to become disengaged from the housing while the release mechanism is depressed and further configured to permit the primary actuator to engage the housing when the release mechanism is released.

16. The method of claim 13, wherein rotating the secondary actuator includes at least one guide member within the secondary actuator engaging with one or more guide sections forming the helical path.

17. The method of claim 13, wherein moving the distal end of the sheath includes releasing a sheath lock and moving a housing portion of a user interface.

18. The method of claim 17, wherein moving the distal end of the sheath includes reengaging the sheath lock once the distal end is adjacent the target region.

19. The method of claim 13, wherein sliding the primary actuator includes depressing an actuator lock to disengage the primary actuator from a housing of a user interface.

20. A method of operating a user interface for positioning electrodes relative to a target region, the method comprising: positioning a distal end of a sheath adjacent to the target region; engaging a primary actuator to move a primary electrode and a secondary electrode collectively within the sheath to extend a distal end of the primary electrode and a distal end of the secondary electrode beyond the distal end of the sheath into the target region; and positioning the secondary electrode by rotating an actuator knob of a secondary actuator to cause the secondary actuator to follow a helical path to move the secondary electrode independently from the primary electrode and extend the distal end of the secondary electrode beyond the distal end of the primary electrode into the target region.

21. The method of claim 20, wherein positioning the distal end of the sheath includes releasing a sheath lock on a sheath actuator to enable movement of a housing relative to a slidable sleeve.
